Experiment 4: Transferring a Block of Data Between Memory Locations Using 8085

AIM:

To write an assembly language program to transfer a block of data from source memory locations to destination memory locations using the 8085 microprocessor.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

ALGORITHM:

1. Initialize the HL pair with the source address 4200H.
2. Initialize the DE pair with the destination address 4300H.
3. Load the number of bytes to be transferred into register C.
4. Load a byte from the source memory location.
5. Transfer the byte to the destination memory location.
6. Increment both source and destination addresses.
7. Decrement the counter and repeat until all bytes are transferred.
8. Stop the program.

PROGRAM:

```
LXI H,4200H

LXI D,4300H

MVI C,05H


LOOP: MOV A,M

STAX D

INX H

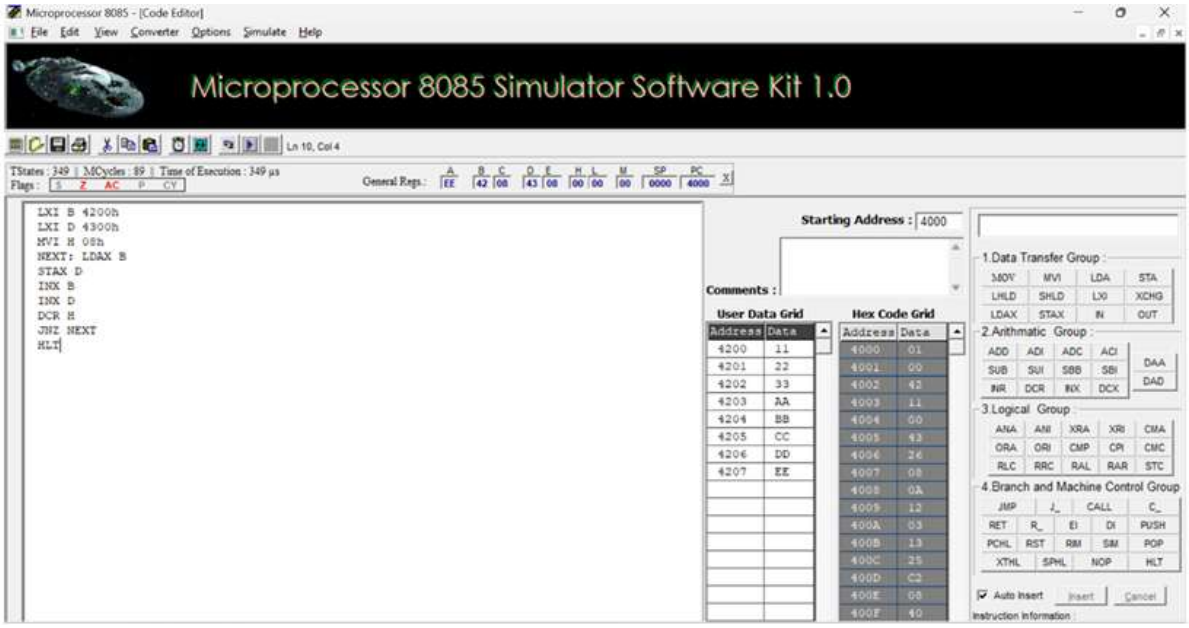
INX D

DCR C

JNZ LOOP


HLT
```

OUTPUT:



RESULT:

Thus, an assembly language program to transfer a block of data between memory locations using the 8085 microprocessor was written and executed successfully.